Cornell CALS
College of Agriculture and Life Sciences

Plant Pathology and Plant-Microbe Biology Section
School of Integrative Plant Science
Cornell Cooperative Extension
Cornell Agri-Tech

# Northern Corn Leaf Blight of Sweet Corn

## Pathogen

Northern corn leaf blight (NCLB) is caused by the fungus, *Exserohilum turcicum*. There are several races of the pathogen and in New York Races 0 and 1 have been detected.

## Hosts

Sweet corn and field corn are the main hosts, but the disease can also affect sorghum, Johnson grass, and sudangrass.

## Significance

NCLB is one of the most prevalent foliar diseases affecting sweet corn in New York. If NCLB causes significant defoliation prior to tassel this may deleteriously impact ear fill, yield, and quality. If defoliation can be delayed until six weeks after silking the disease is unlikely to directly affect yield. However, disease symptoms on the foliage also lead to lesions on the husks which makes them unsuitable for fresh market sales or requiring extra labor for dehusking prior to sale.

## Symptoms

NCLB lesions are typically elliptical (cigar or boat shaped) in shape and large (1 to 6 inches long; Fig. 1A). Lesions on sweet corn cultivars containing resistance genes may appear as smaller chlorotic lesions. Young lesions may appear a grayish-green color and tend to become brown with age due to the fungus producing spores within them, which results in distinct dark areas and a “dirty” appearance (Fig. 1B). These spores are only visible through a hand lens. Lesions first appear on lower leaves after which the infection spreads into the upper canopy. On severely affected plants, almost all leaves may be diseased, and plants may be defoliated. Later in the season, NCLB symptoms may appear similar to other diseases such as Stewart’s Wilt and Goss’s Wilt, and abiotic disorders such as heat stress. Please contact your local CCE office or a diagnostic clinic for assistance with symptom identification.

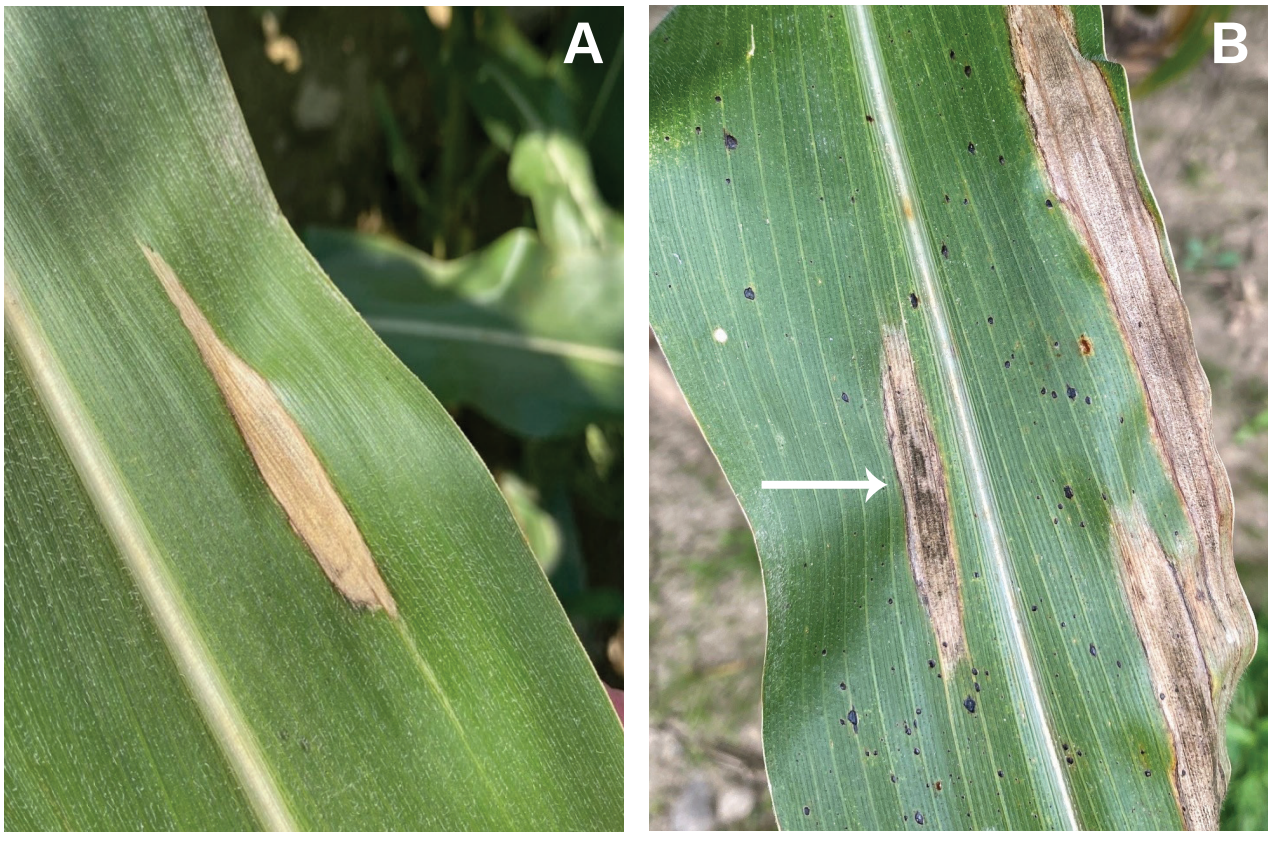


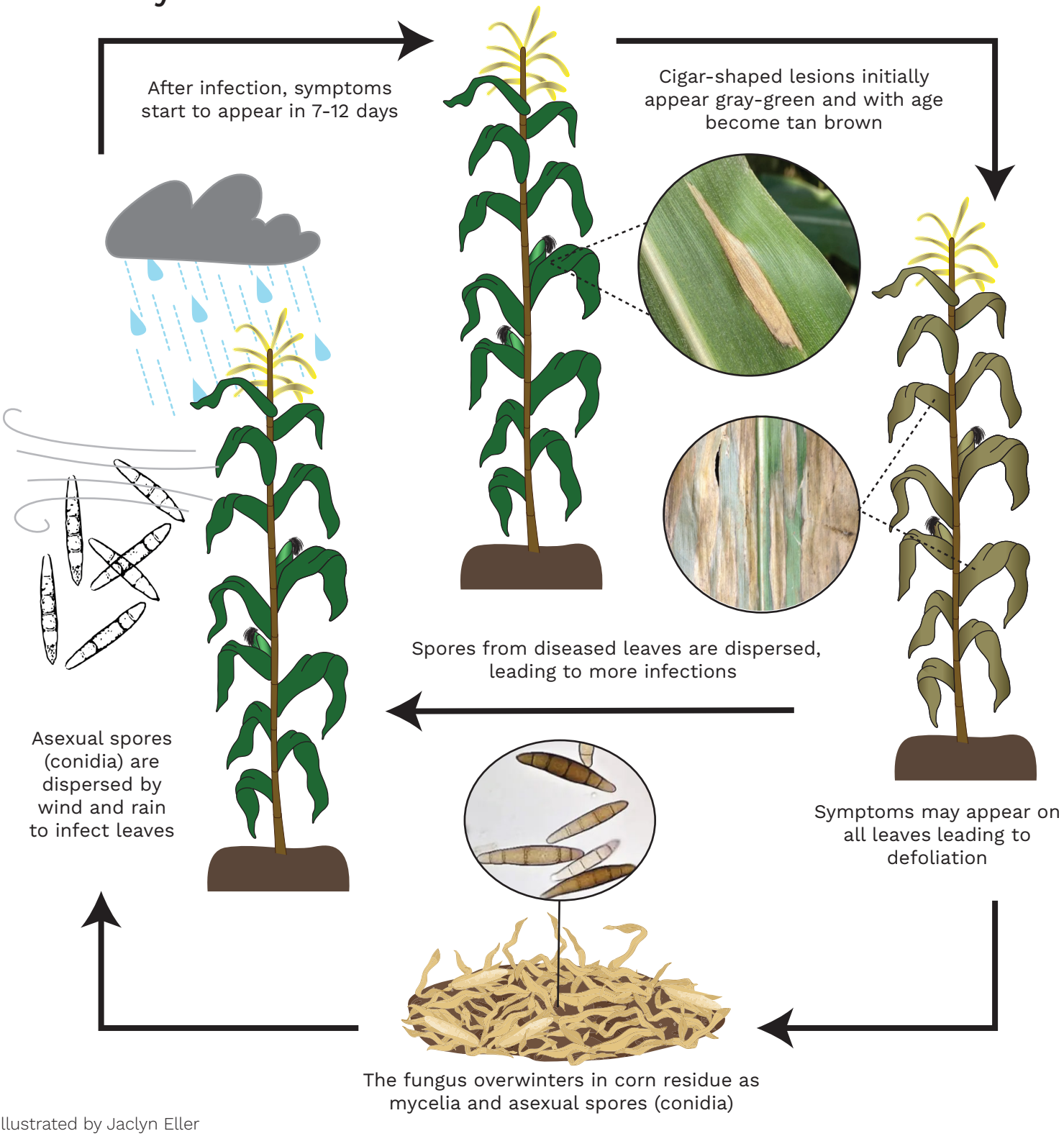

**Figure 1**: (A) NCLB lesions are typically elliptical in shape and are approximately 1 to 6 inches long. Young lesions appear grayish-green color and tend to become brown with age due to spore production within them, giving them a (B) distinct dark “dirty” appearance (arrow).

Prepared by Jaclyn Eller, Julie Kikkert and Sarah Pethybridge (April 2025)

## Disease Development

The fungus, *E. turcicum* overwinters in infested corn residue as conidia and mycelia (Fig. 2). *E. turcicum* can also survive between growing seasons and spread within the season on alternative hosts (Hosts). Infections and disease spread are promoted by warm temperatures (65 to 80˚F) and when leaves are wet to touch for at least 6 hours. The disease is therefore more prevalent in late season sweet corn plantings. After infection, disease symptoms appear within 7 to 12 days. These conditions typically appear in the mid to late growing season leading to multiple infection cycles within the season and rapid disease spread. The lesions on the leaves contain thousands of spores which are dispersed to new leaves, plants and fields by rain splash and wind (Fig. 2).

## Disease Cycle




Illustrated by Jaclyn Eller
Images by Sarah Pethybridge and Leach et al. (1977)

**Figure 2:** Northern corn leaf blight of sweet corn disease cycle.

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# Disease Management

Designing an integrated management strategy for NCLB relies upon the implementation of techniques for breaking the infection cycles or reducing disease progress within the growing season (Fig. 2).

**Inoculum sources**

- *Susceptible Hosts.* Sweet corn, field corn and related species (see Hosts) can be infected by *E. turcicum*, which perpetuates the inoculum within the cropping systems. Due to the large production areas of sweet and field corn across New York, *E. turcicum* is considered endemic and ubiquitous.
- *Seed.* The pathogen, *E. turcicum* is not seed transmitted.
- *Infested Crop Residues.* The pathogen, *E. turcicum* can survive between growing seasons on infested corn residues.

**Management**

**Cultural Practices**

- *Crop Rotation.* Rotating with a non-host species (not field or sweet corn) for at least three years is recommended to minimize the contributions of inoculum from infested crop residue within the field. However, spores of the fungus are also able to be dispersed by wind between fields.
- *Tillage.* Corn residue should be buried by tillage after harvest to encourage decomposition. The fungus cannot survive in the soil without plant residue.
- *Irrigation.* If irrigation is used, applying at midday to minimize the time that leaves are wet will reduce infection and disease spread. Ideally, leaves should not be wet to touch for longer than 6 hours.

**Cultivar Selection**

There are two races of the fungus that cause NCLB in New York, however the dominance of one race over the other is not known. Research is underway to characterize the susceptibility of sweet corn cultivars commonly grown in New York to NCLB.

**Fungicides**

In sweet corn, fungicides are typically applied at V7-V8 and tassel. The aim of applying fungicides for the control of NCLB is to reduce defoliation during reproductive stages of growth and protect the ear leaves. Please check for latest registration updates in the Cornell Vegetable Guidelines.

- *Conventional Fungicides.* Selected conventional fungicides registered for NCLB control in sweet corn in New York are provided in Table 1. There are no cases of fungicide resistance.

**Table 1.** Selected conventional fungicides registered for Northern Corn Leaf Blight control in sweet corn in New York.

| Products | Active Ingrediant(s) | Fungicide Resistance Action Group |
|---|---|---|
| Headline AMP | Metconazole + pyraclostrobin | 3 + 11 |
| Delaro Complete | Prothioconazole + trifloxystrobin + fluopyram | 3 + 11 + 7 |
| Veltyma | Mefentrifluconazole + pyraclostrobin | 3 + 11 |
| Trivapro | Propiconazole + azoxystrobin + benzovindiflupyr | 3 + 11 + 7 |
| Miravis Neo | Propiconazole + azoxystrobin + pydiflumetofen | 3 + 11 + 7 |

*OMRI-listed fungicides.* There are also a broad range of OMRI-listed fungicides registered for NCLB control in sweet corn. However, to date, no efficacy data is available.

# Additional Information

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